

KAARO HEALTH

Cloud Infrastructure Migration Guide

From Offline Clinics to a Connected, Scalable Healthcare Network

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1. Executive Summary

Kaaro Health operates 79+ nurse-led clinics across rural Uganda, delivering affordable healthcare through telemedicine, laboratory services, and medical logistics. The company's current infrastructure — built on offline Clinic Master instances, WhatsApp-based ordering, and paper logs — has served early growth well, but is now a ceiling on scale.

This guide provides Kaaro Health and its technical partners with a structured, practical roadmap for migrating to cloud-based infrastructure. It is grounded in three realities that shape every recommendation:

- Low-resource environment: Clinics operate in areas with intermittent connectivity; solutions must work offline-first and sync when online.
- Operational continuity: Nurses cannot experience downtime during active patient care; migration must be zero-disruption.
- Cost sensitivity: Infrastructure costs must scale proportionally with Kaaro's clinic count, not front-load expense.

Strategic Goal

Replace fragmented, offline-only workflows with a cloud-connected system that gives Kaaro Health HQ real-time visibility into inventory, orders, and clinic operations — while keeping the clinic experience simple, fast, and resilient to poor connectivity.

The recommended approach is a phased migration over approximately 12 months: beginning with cloud foundation and inventory digitization, progressing to structured data integration, and culminating in a scalable ERP layer with EMR readiness. AWS is recommended as the primary cloud provider, with specific services mapped to each workstream.

2. Current State Assessment

2.1 Infrastructure Overview

Understanding the existing technical landscape is the essential starting point for any migration. Kaaro Health's current setup has evolved pragmatically — each component solved an immediate problem — but the resulting architecture lacks integration, visibility, and scalability.

Component	Current State
Patient Records	Clinic Master desktop software on local LAN; no remote access; data siloed per clinic
Inventory Tracking	WhatsApp messages + paper logs; no real-time visibility at HQ
Order Placement	Nurses WhatsApp orders to a coordinator; no structured form or confirmation flow
Clinic Communication	WhatsApp groups; informal, unstructured, difficult to audit
Reporting	Manual spreadsheet aggregation; slow, error-prone, no live data
Connectivity	Intermittent mobile data; no reliable broadband at most clinic sites

2.2 Key Pain Points

- No centralized view of stock levels across 79 clinics; stockouts go undetected until a nurse flags them manually.
- Order aggregation is a full-time manual task; delays in fulfillment directly impact patient care.
- Clinic Master data is inaccessible to HQ without physical presence or manual export.
- Each new clinic requires manual setup, configuration, and onboarding — no standardized deployment pipeline.
- Decision-making (restocking, logistics routing) relies on anecdotal information rather than data.

Constraint Reminder

Every architectural decision in this guide must account for: intermittent connectivity, nurse-facing simplicity, low device capability (feature phones / low-end Android), and tight operational budget. Complexity is the enemy of adoption.

2.3 Migration Readiness Score

The table below scores Kaaro's current readiness across key migration dimensions (1 = not ready, 5 = fully ready). This informs where to invest first.

Dimension	Score	Implication
Connectivity	2 / 5	Offline-first architecture is mandatory
Data Structure	2 / 5	Significant ETL work before cloud ingestion
Staff Digital Literacy	3 / 5	UI must be ultra-simple; training required
Device Availability	3 / 5	Mobile-responsive PWA preferred over native app
Budget Flexibility	3 / 5	Start minimal; use AWS Free Tier + credits
Change Management	4 / 5	Brendah's buy-in established; momentum exists
Leadership Clarity	4 / 5	Product direction being finalized

3. Architecture Principles

Before selecting specific services, the following principles must govern every architectural decision. These are not generic best practices — they are derived directly from Kaaro's constraints.

Principle 1 — Offline-First, Sync-When-Online

The system must function completely at the clinic level without an active internet connection. Data is written locally and synchronized to the cloud when connectivity is available. This is the single most important constraint and must be treated as a hard requirement, not a nice-to-have.

Principle 2 — Progressive Complexity

Do not build the final system on day one. Start with the smallest working version of each component, validate it with real clinic workflows, then expand. Complexity introduced before adoption is waste.

Principle 3 — Mobile-First Interface

Nurses interact primarily via low-end Android smartphones. Every user-facing interface must be designed for mobile-first, with minimal data transfer, fast load on 3G, and large touch targets. Progressive Web Apps (PWAs) are strongly preferred over native applications.

Principle 4 — Single Source of Truth

All inventory, order, and patient data must ultimately resolve to one authoritative record in the cloud. Conflicts from offline sync must be handled gracefully with clear resolution rules (last-write or version-vector, depending on data type).

Principle 5 — Cost-Proportional Scaling

Infrastructure cost must scale with the number of active clinics, not with a fixed baseline. Serverless and managed services should be preferred where they eliminate idle resource costs. Budget should never constrain care delivery.

Principle 6 — Least Privilege Security

Each clinic, user role, and service should have access only to the data it needs. Clinic nurses see their clinic data; HQ coordinators see aggregate data; no role has unfiltered access to the full dataset. Patient data must be encrypted at rest and in transit at all times.

Principle 7 — Incremental Migration (No Big Bang)

Existing Clinic Master workflows must continue operating throughout the migration. New cloud systems run in parallel, with clinics migrating one by one. No clinic goes offline during a migration event.

Industry Best Practice — CAP Theorem Applied

In distributed systems with unreliable connectivity, you must choose between Consistency and Availability. For Kaaro's inventory system, Availability wins: a nurse must always be able to record a dispensing event, even offline. Eventual consistency is acceptable. For financial data (payments, invoices), Consistency wins and those writes must queue until connectivity is restored.

4. Recommended AWS Architecture

4.1 Why AWS

Amazon Web Services is recommended as Kaaro's primary cloud provider for the following reasons:

- **AWS Activate:** Startups in global health and social impact sectors can access \$5,000–\$100,000 in AWS credits, significantly reducing initial cost.
- **Africa region availability:** AWS has infrastructure in Cape Town (af-south-1) providing lower latency for East African users compared to European or US regions.
- **Mature managed services:** AWS offers production-grade managed services for every component of this architecture, reducing the need for DevOps expertise that Kaaro does not currently have in-house.
- **Ecosystem breadth:** AWS Amplify, AppSync, and DynamoDB Global Tables are particularly well-suited to offline-first mobile applications — a direct match for Kaaro's primary constraint.

Criteria	AWS (Recommended)	GCP	Azure
Offline-first sync (built-in)	✓ AppSync DataStore	✗ Manual build required	✗ Manual build required
Startup credits available	✓ Up to \$100,000	✓ Up to \$200,000	✓ Up to \$150,000
Nearest Africa region	✓ Cape Town (af-south-1)	✓ Johannesburg	✓ South Africa North
Mobile PWA / offline stack	✓ Amplify (production-ready)	⚠ Firebase (limited offline)	⚠ No managed solution
Managed serverless database	✓ DynamoDB	✓ Firestore	✓ Cosmos DB
Ease of use for small team	✓ High	⚠ Medium	⚠ Medium
Key differentiator	Best offline-sync stack	Best ML/analytics (BigQuery)	Best if Microsoft ecosystem

Action Item

Apply for AWS Activate Founders (\$1,000 in credits, no AWS equity required) immediately. If Kaaro is associated with an accelerator or NGO partner, apply for the Portfolio tier (\$10,000–\$100,000). This should happen before any architecture work begins.

4.2 Core Services Map

Workstream	AWS Service	Purpose for Kaaro
Offline Sync	AWS AppSync + DynamoDB	GraphQL API with built-in conflict resolution; DynamoDB stores clinic-level records with eventual consistency sync
Inventory DB	Amazon DynamoDB	NoSQL, serverless, scales from 1 to 1000 clinics with zero configuration changes; per-request billing
File Storage	Amazon S3	Store paper-log scans, prescription images, and backup exports; lifecycle policies auto-archive old data cheaply
Analytics / HQ Dashboard	Amazon QuickSight	Business intelligence layer; connects to DynamoDB via Athena; no SQL expertise required for nurses
Data Warehouse	Amazon Athena + S3	Query structured exports (CSV/JSON) from Clinic Master during transition; serverless, pay-per-query
Authentication	Amazon Cognito	User identity for nurses, coordinators, HQ admin; supports offline tokens; free for first 50,000 MAUs
API Gateway	Amazon API Gateway	RESTful API layer between mobile PWA and backend; rate limiting, auth, and request routing
Serverless Compute	AWS Lambda	Event-driven logic (e.g., trigger reorder alert when stock < threshold); no servers to manage
Notifications	Amazon SNS / SES	Push low-stock alerts to clinic nurses via SMS or WhatsApp Business API integration
Infrastructure as Code	AWS CDK / CloudFormation	Define entire infrastructure in code; deploy a new clinic's cloud environment in minutes, not days
Monitoring	Amazon CloudWatch	Metrics, alarms, and logs for all services; detect sync failures or API errors before users notice
Secrets Management	AWS Secrets Manager	Store API keys, DB credentials, and integration tokens securely; never hardcode credentials

4.3 Offline-First Data Architecture (Deep Dive)

This is the highest-risk component of the migration and deserves detailed treatment. The recommended pattern is:

1. Clinic-level PWA stores all writes to a local IndexedDB (browser-native, works offline).
2. On connectivity restore, AppSync's built-in offline mutation queue replays pending writes to DynamoDB.

3. Conflict resolution rule: for inventory dispensing events, last-write-wins by timestamp. For order confirmations, server-side wins (orders must be confirmed by HQ).
4. DynamoDB Streams capture all changes and trigger Lambda functions for downstream processing (alerts, analytics, reporting).
5. HQ dashboard reads from DynamoDB via AppSync and renders in near real-time.

Industry Best Practice — Idempotent Writes

Every write operation from the clinic must carry a unique client-generated ID (UUID). If the same write replays due to a sync retry, the backend detects the duplicate ID and ignores the second write. Without idempotency, a nurse dispensing 5 units offline could record 10 or 15 units after reconnection. This is non-negotiable for inventory accuracy.

5. Phased Migration Roadmap

The migration is structured across three phases, each delivering standalone value. Kaaro can stop after any phase and still have a meaningfully better system than today.

PHASE 1 | Foundation & Inventory Digitization | Months 1–4

Goal: Replace WhatsApp-based inventory ordering with a structured, cloud-backed system. Establish cloud account, security baseline, and data pipeline from real clinic data.

Phase 1 Deliverables

- AWS account created, IAM structure defined, billing alerts configured.
- AWS Activate credits applied (do this first).
- DynamoDB schema designed for clinics, medicines, and inventory_records tables.
- Inventory dashboard PWA deployed with real Kaaro clinic + medicine data.
- Offline sync via AppSync operational for at least 5 pilot clinics.
- Low-stock alert via SNS/SMS functional.
- Order placement flow: nurse submits → HQ receives structured order → confirmation sent back.

Phase 1 Technical Steps

#	Task	Detail
1	AWS Account Setup	Create AWS Organization with root account + separate accounts for dev, staging, production. Enable CloudTrail from day one.
2	IAM Roles & Policies	Define roles: KaaroAdmin, KaaroHQCoordinator, KaaroClinicNurse. Apply least-privilege policies. Enable MFA on root and admin accounts.
3	Billing Guardrails	Set monthly budget alert at \$50 in AWS Budgets. Configure Cost Explorer. Tag all resources with project:kaaro-health and environment:prod.
4	DynamoDB Schema	Design tables: Clinics, Medicines, InventoryLedger (append-only), Orders, OrderItems. Use clinic_id as partition key throughout for data isolation.
5	Cognito User Pool	Create nurse and coordinator user groups. Implement email/phone login. Issue short-lived JWTs for API authorization.
6	AppSync API	Define GraphQL schema. Configure DynamoDB resolvers. Enable DataStore for offline mutation queue and conflict resolution.
7	Data Migration Script	Extract clinic list and medicine catalog from existing Kaaro data (spreadsheets / Clinic Master exports). Validate and seed into DynamoDB.
8	PWA Deployment	Deploy inventory dashboard to AWS Amplify Hosting. Configure service worker for offline caching. Test on low-end Android devices on 3G.

#	Task	Detail
9	Pilot Rollout	Select 5 clinics with engaged nurses. Run parallel with WhatsApp for 4 weeks. Compare order accuracy and fulfillment time.
10	Monitoring Setup	CloudWatch dashboards for sync error rate, order latency, and API error rate. PagerDuty or SNS alerts for critical failures.

PHASE 2 | Data Integration & Operational Visibility | Months 5–8

Goal: Connect the cloud layer to existing Clinic Master data. Build HQ analytics and reporting. Expand rollout to all 79 clinics.

Phase 2 Deliverables

- Clinic Master data export pipeline: scheduled ETL to S3, queryable via Athena.
- HQ QuickSight dashboards for inventory trends, consumption patterns, and reorder recommendations.
- Automated demand forecasting model (simple moving average to start) running on Lambda.
- All 79 clinics migrated to PWA order flow.
- WhatsApp ordering officially deprecated (parallel period ends).
- Kaaro internal team trained on cloud console basics and dashboard interpretation.

Phase 2 Technical Steps

- Build ETL pipeline: Clinic Master CSV export → S3 → Glue Crawler → Athena table → QuickSight dataset.
- Implement scheduled Lambda to compute 30/60/90-day consumption trends per medicine per clinic.
- Design reorder recommendation engine: if (current_stock / avg_daily_consumption) < reorder_lead_time, flag.
- Deploy clinic onboarding automation with CDK: new clinic = run cdk deploy, all resources provisioned in minutes.
- Implement audit trail: every inventory write event logged immutably to S3 with DynamoDB Streams + Firehose.
- Conduct usability testing across 3 clinic archetypes (urban, peri-urban, rural-remote); adjust UI based on findings.

PHASE 3 | ERP Foundation & EMR Readiness | Months 9–12

Goal: Build the scalable ERP layer for supply chain management, financial tracking, and EMR-readiness. Position Kaaro for 200+ clinic scale.

Phase 3 Deliverables

- Structured supply chain module: procurement, vendor management, delivery tracking.

- Financial reconciliation layer: clinic revenue, rent-to-own payments, and logistics charges tracked in cloud.
- API-ready data model that a future EMR (e.g., OpenMRS, Bahmni) can connect to.
- Multi-region backup strategy with RTO < 4 hours and RPO < 1 hour.
- Security audit and penetration test before patient data is stored in cloud.

Phase 3 Dependency Note

Phase 3 EMR work should not begin until Kaaro has confirmed their EMR selection and conducted a HIPAA/local data residency compliance review. Uganda's National Health Data Policy may impose local data storage requirements. This must be verified with legal counsel before patient records are moved to cloud.

6. Data Migration Strategy

6.1 Data Inventory

Before migrating any data, conduct a complete audit of existing data assets. The table below maps known Kaaro data sources to their migration approach.

Data Source	Format	Priority	Migration Approach
Clinic List	Spreadsheet	P0 — Week 1	Manual clean + seed script into DynamoDB Clinics table
Medicine Catalog	Spreadsheet / paper	P0 — Week 1	Standardize names/units; assign medicine_id; seed into Medicines table
Current Inventory	WhatsApp + paper	P0 — Week 2	Point-in-time count at each pilot clinic; nurse-verified; load as opening balance
Historical Orders	WhatsApp export + spreadsheets	P1 — Month 2	Parse WhatsApp export (Python script); map to structured schema; load to S3 for Athena queries (not DynamoDB)
Patient Records (Clinic Master)	Local SQL DB	P2 — Phase 2	Scheduled CSV export → S3 → Athena; no migration of live patient records until EMR decision
Financial Records	Spreadsheets	P2 — Phase 3	Structured import once financial module is scoped

6.2 Data Quality Rules

Raw data from WhatsApp logs and paper records will be dirty. Define validation rules before loading:

- Medicine names: normalize to a canonical name list (e.g., 'amoxicillin 500mg' = 'Amoxicillin 500mg caps'); reject unknowns.
- Quantities: must be non-negative integers; fractional values flagged for nurse review.
- Clinic IDs: all records must reference a valid clinic_id from the Clinics master table; orphaned records quarantined.
- Dates: standardize to ISO 8601 (YYYY-MM-DD); ambiguous dates flagged and held for manual resolution.

Industry Best Practice — Data Quarantine Pattern

Never reject bad data silently. All records that fail validation go into a quarantine table in S3 with a rejection_reason tag. A coordinator reviews the quarantine table weekly and either corrects the record or marks it as invalid. This prevents data loss while maintaining data quality.

7. Security & Compliance

7.1 Security Baseline (Non-Negotiable from Day 1)

Healthcare data requires stringent security controls. The following must be in place before any patient data touches cloud infrastructure.

Control	Implementation
Encryption at rest	All DynamoDB tables and S3 buckets encrypted with AWS KMS customer-managed keys. Enable by default on every resource.
Encryption in transit	All API calls over HTTPS/TLS 1.2+. AppSync enforces HTTPS. Reject HTTP on API Gateway.
Multi-Factor Authentication	Mandatory for all AWS console access. Cognito user pool requires MFA for coordinator and admin roles.
Least privilege IAM	No wildcard (*) permissions in any IAM policy. Use AWS IAM Access Analyzer to detect overpermissioned roles.
VPC Isolation	DynamoDB accessed via VPC endpoint (not public internet). Lambda functions run inside VPC.
Audit logging	CloudTrail enabled in all regions. API Gateway access logs enabled. S3 server access logging enabled.
Secrets Management	Zero hardcoded credentials anywhere. All secrets in AWS Secrets Manager with auto-rotation enabled.
Backup & Recovery	DynamoDB point-in-time recovery (PITR) enabled. S3 versioning enabled. Test restore procedure quarterly.

7.2 Compliance Considerations

Kaaro operates in Uganda. The following regulatory context must be reviewed with legal counsel:

- Uganda Data Protection and Privacy Act (2019): Governs collection, processing, and storage of personal data. Cloud provider must be disclosed to data subjects.
- National Health Data Policy: May require certain patient data categories to remain within Uganda's borders. Verify before enabling af-south-1 (Cape Town) as sole region for PHI.
- If Kaaro seeks international partnerships or funding, HIPAA technical safeguard alignment is recommended even if not legally required in Uganda — it signals maturity to partners.

Recommendation
Engage a local Ugandan legal advisor to review data residency requirements before Phase 3 begins. In the meantime, keep all patient data in Clinic Master (offline) and limit cloud to inventory, orders, and operational data — none of which is PHI.

8. Cost Estimation

Cost estimates below are based on 79 clinics, moderate usage, and AWS af-south-1 pricing as of early 2026. Actual costs will vary; use AWS Pricing Calculator for live estimates before committing.

Service	Est. Monthly Cost	Notes
DynamoDB (on-demand)	~\$8–15	Low read/write volume for 79 clinics; scales automatically
AppSync	~\$4–8	GraphQL requests + real-time subscriptions for HQ dashboard
AWS Lambda	~\$1–3	Generous free tier; alerts and processing logic well within it
S3	~\$2–5	Historical data, exports, backups; lifecycle to Glacier after 90 days
Cognito	\$0	Free for first 50,000 MAUs; 79 clinics + HQ well within limit
CloudWatch	~\$3–6	Logs, metrics, alarms across all services
QuickSight (Phase 2)	~\$18–24	\$9/author/month; 2 HQ authors + reader sessions
Amplify Hosting	~\$2–5	PWA hosting; includes CDN and CI/CD pipeline
Athena (Phase 2)	~\$2–5	Pay per query on historical data; minimal if queries are optimized
API Gateway	~\$2–4	REST API calls; first 1M requests/month free
TOTAL ESTIMATE	~\$42–75/month	Phase 1+2 fully operational. AWS credits offset 6–12+ months.

At Kaaro's scale (79 clinics, ~\$330/month revenue per clinic), this infrastructure cost represents less than 0.3% of monthly revenue from clinic rent-to-own alone — a strongly positive unit economics case for digital investment.

9. Change Management & Training

Technology migration succeeds or fails based on adoption, not architecture. Kaaro's greatest risk is not technical — it is nurses abandoning the new system and reverting to WhatsApp because the new interface is confusing or unreliable.

9.1 Adoption Strategy

- **Champion clinic model:** Identify 2–3 nurses who are enthusiastic early adopters. Give them early access, collect feedback, make them internal advocates before broader rollout.
- **Parallel running:** New system and WhatsApp ordering run simultaneously for 4 weeks per clinic cohort. Nurses are not forced off until they express confidence.
- **Feedback loop:** Weekly 15-minute check-in calls between HQ coordinator and pilot clinic nurses. Track friction points and fix before the next cohort.
- **Incentive alignment:** Frame the new system as making nurses' jobs easier, not adding monitoring. Show nurses their own consumption data and how it prevents stockouts they've experienced.

9.2 Training Plan

Audience	Format	Content
Clinic Nurses	30-min in-person demo + WhatsApp video tutorial	How to check stock levels, submit an order, read a low-stock alert. No technical content.
HQ Coordinators	2-hr workshop	How to use the HQ dashboard, process incoming orders, interpret consumption trends, flag anomalies.
Kaaro IT / Admin	4-hr technical session	AWS console basics, how to add a new clinic, how to read CloudWatch alarms, escalation path for outages.
Leadership (Brendah)	1-hr executive briefing	Dashboard KPIs, cost visibility, migration progress tracking, strategic roadmap view.

10. Risk Register

Proactively identifying and planning for risks is a core discipline of cloud migrations. The following risks are specific to Kaaro's context.

Risk	Likelihood	Impact	Mitigation
Connectivity worse than assumed; sync fails routinely	High	High	Offline-first architecture mandatory. Test on 2G simulation. Exponential backoff on sync retry. Never block nurse workflow on connectivity.
Nurses reject new UI; revert to WhatsApp	Medium	High	Champion model + parallel running. Simplicity-first design. Never remove WhatsApp until nurses request it be removed.
Data quality too poor to migrate meaningfully	Medium	Medium	Data quarantine pattern. Start with forward data (new records only) if historical data is too dirty. Historical data is analytics-only, not operational.
AWS costs exceed budget due to misconfiguration	Low	High	Billing alerts from day one. AWS Budgets with hard threshold. Monthly cost review. DynamoDB on-demand billing prevents surprise spikes.
Data residency compliance violation	Low	High	Keep PHI offline until legal review complete. Cloud stores only inventory and operational data in Phase 1-2.
Key team member turnover mid-migration	Medium	Medium	Infrastructure as Code (CDK) ensures all architecture is documented in code, not in someone's head. Runbook documentation required for each phase.
Clinic Master vendor locks data format	Low	Medium	Request CSV/SQL export capability in writing. Test export script early in Phase 1. Do not depend on vendor API.
AWS region outage (af-south-1)	Very Low	Medium	Enable DynamoDB global tables (us-east-1 as secondary) for Phase 3 only. For Phase 1-2, accept single-region risk given cost constraints.

11. Immediate Next Steps

The following actions should be taken in the next 30 days to begin the migration with momentum.

#	Action	Detail	Owner	Timeline
1	Apply for AWS Activate	Register at aws.amazon.com/activate . Apply for Founders tier minimum. Unlock credits before any paid resources.	Consulting Team	Week 1
2	Confirm Product Direction	Decision: analytics dashboard vs lightweight ERP vs full ERP. Required before schema design.	Kaaro Leadership	Week 1
3	Export Clinic & Medicine Master Data	Get complete clinic list + medicine catalog from Kaaro in any format (spreadsheet, export). This is the seed data.	Kaaro HQ	Week 1–2
4	Create AWS Account	Set up AWS Organization. Enable CloudTrail. Set billing alert at \$50/month. Tag everything.	Consulting Team	Week 1
5	Select 5 Pilot Clinics	Choose clinics with engaged nurses and semi-reliable connectivity. Get Brendah's approval.	Kaaro + Consulting	Week 2
6	Design DynamoDB Schema	Finalize table design based on confirmed product direction. Review with Kaaro coordinator.	Consulting Team	Week 2–3
7	Build PWA Prototype v2 with Real Data	Replace mock data in current prototype with actual clinic + medicine records.	Consulting Team	Week 3–4
8	Pilot Launch	Deploy to 5 clinics. Run parallel with WhatsApp. Start collecting real usage data.	Consulting + Kaaro	Week 4–5